

Examination Question Paper

e-DBDA September 2020

Module Name: Advanced Analytics

DATE: 27-02-2021

Duration: 2hrs

Max. Marks: 40

Instructions for the exam:

1. All questions are compulsory
2. Make sure to attempt all sub questions
3. Your analysis should have justifications or explanation (in English)
4. Before creating Zip file of your .ipynb files make sure that you have closed all .ipynb files.

Instructions for Uploading:

- Students should upload only zip file
- Kindly ensure before creating zip file, all the required files should be closed to avoid any corruption of the zip file
- Please make sure all file/folders are copied to the main folder before making zip file
- Zip File should be renamed with FULL PRN No. of the candidate.
(For example: 200940183001-Yourfullname.zip)
- Uploading is allowed only once

Q1. Apply Analytical Framework for comparing classifiers. Using suitable cost matrix for comparing 3 different classifiers on given data. **(20 marks)**

[Note: Keep data cleaning to minimum. Classifiers can be fine-tuned as per time availability.]

Data Link : <https://www.kaggle.com/divyansh22/crystal-system-properties-for-liion-batteries>

[5 Marks for each classifier results + 5 marks for Analytical framework output]

Q2. Do Time Series prediction on following Data. Use ARIMA model. **(20 Marks)**

Select any one currency

1. Make the data stationary using at least 2 ways (8 marks - 4 marks each method)
 2. Visualize TS components (4 marks)
 3. Prediction using ARIMA with two different values of p,d,q
- Compare the two ARIMA models based on RMSE and MAPE(8 marks)

Data Link : <https://www.kaggle.com/lsind18/euro-exchange-daily-rates-19992020>

*****All the best*****